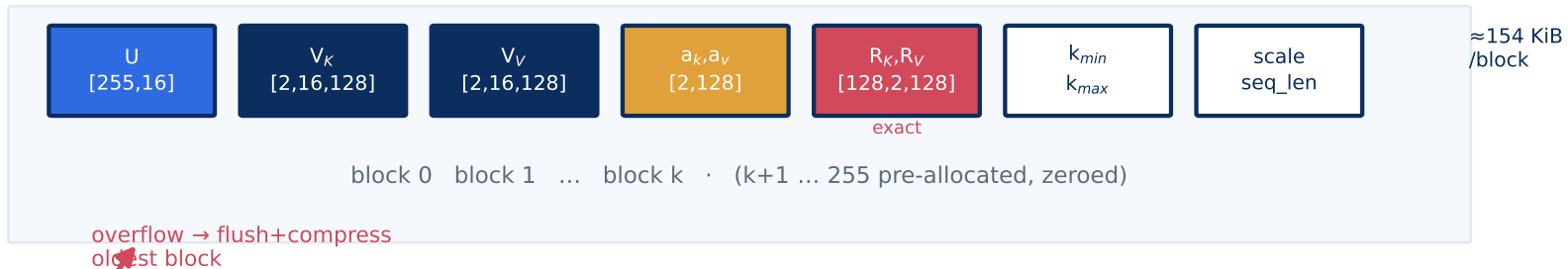


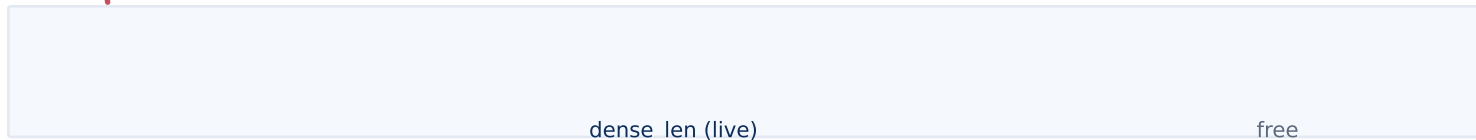
Per-Session KV Store — memory layout (one layer of 28)

Bounded dense recency window + fixed compressed pool — the state that does NOT grow unboundedly with context

Compressed pool (fixed $M = 256$ blocks · $\leq 65\,536$ tokens)



Dense recency window (sliding, capacity $W+B = 768$ tokens, uncompressed fp16)



Footprint per layer $\approx$ dense $768 \cdot H_{kv} \cdot d \cdot 2 \cdot 2$ B + used_blocks · 154 KiB.

The dense window is constant; the compressed term grows at $\approx 1/1.66$ the dense-KV rate (default R), and the pool is capped at 256 blocks → the bounded memory slope.